

Problem Set 8
Due: Apr. 26, 2021

1. Section 4.1 Problem 4.

The metro train arrives at the station near your home every quarter hour starting at 6:00 a.m. You walk into the station every morning between 7:10 and 7:30 a.m., with the time in this interval being a random variable with given PDF (cf. Example 3.14, in Chapter 3). Let X be the elapsed time, in minutes, between 7:10 and the time of your arrival. Let Y be the time that you have to wait until you board a train. Calculate the CDF of Y in terms of the CDF of X and differentiate to obtain a formula for the PDF of Y .

2. Section 4.1 Problem 9.

Consider the same problem as in Example 4.9, but assume that the random variables X and Y are independent and exponentially distributed with different parameters λ and μ , respectively. Find the PDF of $X - Y$.

3. Find the probability density function of $Y = e^X$ when X is normally distributed with parameters μ and σ^2 . The random variable Y is said to have a *lognormal distribution* (since $\log Y$ has a normal distribution) with parameters μ and σ^2 .
4. A point (X, Y) is picked at random uniformly in the unit circle. Find the joint PDF of R and X , where $R^2 = X^2 + Y^2$.